



Effect of chronic treatment with *Enicostemma littorale* in non-insulin-dependent diabetic (NIDDM) rats

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Abstract

We have studied the effect of aqueous extract of *Enicostemma littorale* (2 g/kg p.o.) daily for 6 weeks in neonatal non-insulin-dependent diabetes mellitus (NIDDM) rats. To induce NIDDM a single dose injection of STZ (70 mg/kg; i.p.) was given to the 5-day-old pups. After 3 months of STZ injection when animals were confirmed as diabetic, *E. littorale* was administered for 6 weeks. Fasting and fed glucose and insulin levels in NIDDM were significantly ($P < 0.05$) higher than control rats and they were significantly decreased by the treatment with *E. littorale*. Results of the oral glucose tolerance test (OGTT) showed a significant ($P < 0.05$) decrease in both AUC_{glucose} and AUC_{insulin} values in NIDDM treated group. Insulin sensitivity (K_{ITT}) index of NIDDM control was significantly lower as compared with Wistar control and this was significantly ($P < 0.05$) increased after treatment with *E. littorale*. Treatment with *E. littorale* also decreased the elevated cholesterol, triglyceride and creatinine levels observed in NIDDM rats. Our data suggest that aqueous extract of *E. littorale* is a potent herbal antidiabetic. It produces an increase in insulin sensitivity, normalizes dyslipidaemia and provides nephroprotection in diabetic rats. © 2002 Elsevier Science Ireland Ltd. All rights reserved.

Keywords: *Enicostemma littorale* (Gentianaceae); Streptozotocin; Insulin resistance

1. Introduction

A number of indigenous plants have been widely used for the treatment of diabetes and several other ailments (Mukerji, 1957). Plants like *Azadirachta indica*, *Eugenia jambolana*, *Momordica charantia*, *Swertia chirata*, *Gymnema sylvestre* etc. have shown a great potential as antidiabetics and they have been included in various polyherbal preparations. (Dhar et al., 1968; Sharma and Jain, 1960; Dhavan et al., 1980; Hakim et al., 1996).

Enicostemma littorale (Gentianaceae) is a glabrous perennial herb belonging to the family Gentianaceae (Kirtikar and Basu, 1935). It grows throughout India up to 1.5 feet height and more frequently near the sea. It is called as Chota-kirayat or Chota chirayata in Hindi, Mamejavo in Gujarati, Nagajivha in Bengal and Vellarugu or Vallari in Tamil.

Dymock et al. (1893) reported that the aerial part of the plant gave 34% of dry alcoholic extract and 15.7% of ash. Qualitative analysis of the ash revealed the presence of minerals like iron, potassium, sodium, calcium, magnesium, silica, phosphate, chloride, sulphate and carbonate. Five alkaloids, two sterols and volatile oil, have been reported by Natarajan and Prasad (1972). Monoterpene alkaloids like-enicoflavin and gentiocrucine were also isolated (Ghosal et al., 1974; Chaudhuri et al., 1975). The presence of catechins, saponins, steroids and triterpenoids were reported by earlier workers (Retnam and DeBritto, 1988). Betulin, a triterpene sapogenin (m.p. 252–254 °C) was also isolated by earlier workers (Rai and Thakar, 1966; Desai et al., 1966).

Swertiamarin (m.p. 108–111 °C) was isolated from the green viscous mass obtained from an alcoholic extract of the drug treated with ether followed by ethylacetate. (Rai and Thakar, 1966; Desai et al., 1966). Flavonoids and xanthenes were found to be present in this plant. Six phenolic acids viz: vanillic acid,

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syringic acid, *p*-hydroxy benzoic acid, protocatechuic acid, *p*-coumaric acid and ferulic acid were also found (Daniel and Sabnis, 1978). Seven flavonoids were isolated from alcoholic extract and their structures were identified as apigenin, genkwanin, isovitexin, swertisin, saponarin, 5-*O*-glucosylswertisin and 5-*O*-glucosylisowertisin (Ghosal and Jaiswal, 1980). Methanol extract of *E. littorale* was found to contain different amino acids like L-glutamic acid, tryptophane, alanine, serine, aspartic acid, L-proline, L-tyrosine, threonine, phenyl alanine, L-histidine monohydrochloride, methionine, iso leucine, L-arginine monohydrochloride, DOPA, L-Glycine, 2-amino butyric acid and valine (Retnam and DeBritto, 1988).

This plant has been used as folk medicine for the treatment of diabetes mellitus in Western and Southern India (Gupta et al., 1962). Various Ayurvedic formulations containing *E. littorale* as one of the ingredients have been shown to produce antihyperglycemic activity in hyperglycemic rat models (Gupta et al., 1962). Ethnomedical studies of North Gujarat (India) reveal the use of hot aqueous extract of *E. littorale* by the tribal inhabitants for the treatment of diabetes, fever, stomach ache, dyspepsia and malaria in interior part of Gujarat. Although the crude extract of *E. littorale* has been used in diabetes mellitus by tribal inhabitants in India, a detailed study for antidiabetic property, insulin sensitivity and diabetes induced metabolic alterations is not available. Hence, the present investigation was undertaken to study the effects of chronic treatment with *E. littorale* on insulin sensitivity and glucose intolerance in STZ-induced NIDDM model of rat. Attempts were made to study the effect of these agents on lipid profile and kidney functions.

2. Materials and methods

2.1. Identification and collection of the plant

E. littorale growing at various places of Gujarat (India) was collected in the month of August and September, at the end of the flowering season. The collection was made by the uprooting method and thereby taking care to collect root as well. The plant was identified by comparing its morphologically and microscopically as mentioned in the different standard books and floras (Kirtikar & Basu, 1935). Professor O.P. Saxena, Ph.D., Head of the Botany Department, Gujarat University, Ahmedabad, India, authenticated its identity. The plant was cleaned and dried in the shade and packed into loose masses in plastic bags. House specimen was deposited at the Botany Department, Gujarat University, Ahmedabad, India.

2.2. Preparation of aqueous extract of *Enicostemma littorale*

Three kilograms of the shade-dried herb containing all vegetative and reproductive parts of *E. littorale* were mixed with 24 l of water and heated till it was reduced to half. The whole mass was filtered and concentrated to further half twice (6 l) by heating. The marc obtained from this was again mixed with fresh 12 l of water and heated till approximately 6 l of water left behind. The whole mass was filtered again and the filtrate was concentrated to 3 l by heating. This time the marc remaining was discarded. Both filtrates were mixed and further concentrated at low temperature. The concentration of dried *E. littorale* was 2 g/ml in the final extract. A 2% sodium benzoate solution was added as a preservative. The extract was stored at a temperature of 15–20 °C in well-closed glass container.

2.3. NIDDM rat model and treatment protocol

Healthy albino rats of Wistar strain were kept for breeding. To induce NIDDM, STZ (70 mg/kg) was administered intraperitoneally to a group of 5-day-old pups. Another group of pups received only saline. The pups were weaned after 30 days, and 3 months after the injection of STZ, the animals were checked for fasting glucose levels. The animals showing fasting glucose levels >120 mg/dl were considered as diabetic. Pups that received saline were considered as control animals. The experimental animals were then divided into the following four groups, (1) Control, (2) Control treated with *E. littorale*, (3) NIDDM control, and (4) NIDDM treated with *E. littorale* (2 g/kg body weight, p.o.). Treatment was given daily for 6 weeks. The control group received an equal volume of the vehicle. During the study standard food and water were provided ad libitum. Changes in body weight, food intake and water intake were recorded.

2.4. Oral glucose tolerance test (OGTT)

At the end of the 6 weeks study, 2 days after recording the blood pressure, rats were subjected to an oral glucose tolerance test (OGTT). Glucose (1.5 g/kg) was administered to 18 h fasted rats. Blood samples were collected from tail vein at 0, 10, 20, 30, 60 and 120 min. Serum was analyzed for glucose by the GOD–POD method by using diagnostic reagent kits (Bayer, Vadodara, India. Ltd.) and insulin by RIA kits (Bhabha Atomic Research Center, Bombay, India). The results were expressed as integrated area under the curve (AUC) for glucose and insulin, that was calculated by the trapezoid rule [$AUC = C_1 + C_2/2X(t_1 - t_2)$] and changes in glucose and insulin concentrations during

OGTT were expressed as AUC_{glucose} (mg/dl per min) and AUC_{insulin} ($\mu\text{U/ml}$ per min), respectively.

2.5. Insulin tolerance test (K_{ITT})

Insulin tolerance test (ITT) (Alford et al., 1971) is used to assess peripheral insulin resistance (IR). This test measures insulin sensitivity using K_{ITT} as an index of insulin mediated glucose metabolism. Rats were fasted for 15 h before giving insulin challenge. Neutral insulin injection (Actrapid Novo, Ahmedabad, India) was diluted with 0.9% saline to a final concentration of 0.2 U/0.1 ml, and then administered in the dose of 0.2 U/100 g body weight by slow i.v injection through tail vein. Blood samples were collected at 0 min and then at 10, 20, 30 and 60 min after administration of insulin injection. Serum was separated and subjected to glucose estimation. K_{ITT} was determined from the slope of a linear portion of the regression line of natural logarithm of glucose versus time (Alford et al., 1971) and using a formula (Lundbaek, 1962), K_{ITT} was calculated as follows,

$$K_{\text{ITT}} = \frac{0.693 \times 100}{t_{1/2}}$$

where $t_{1/2}$ represents the half-life of plasma glucose decay. The half-life of plasma glucose was obtained by plotting plasma glucose concentrations versus time on semilogarithmic graph paper.

2.6. Statistical analysis

Results were analyzed statistically using analysis of variance (ANOVA) followed by Tukey's test. A level of 5% ($P < 0.05$) was taken as the level of statistical significance.

3. Results

3.1. Effect on general parameters

Animals which received STZ showed a significant reduction in weight gain as compared with the control animals ($P < 0.05$). Treatment with *E. littorale* caused slight but significant increase in the weight gain in the body weight of these animals (Table 1).

3.2. Effect on glucose and insulin levels

Both fasting as well as fed glucose levels were significantly ($P < 0.05$) higher in NIDDM animals as compared with those in control animals (Fig. 1); this was associated with hyperinsulinemia in NIDDM control animals (Fig. 2). Treatment with *E. littorale*

significantly prevented the rise in fed glucose level as well as insulin levels.

Both AUC_{glucose} and AUC_{insulin} were significantly higher in NIDDM control as compared with control animals ($P < 0.05$) and treatment with *E. littorale* significantly decreased both AUC_{glucose} and AUC_{insulin} values (Figs. 3 and 4) and prevented decrease in K_{ITT} value in NIDDM rats. In all other groups it did not alter K_{ITT} value (Table 2).

3.3. Effect on serum lipids and kidney functions

STZ induced NIDDM rats showed significant hypercholesterolemia as compared with control. Treatment with *E. littorale* showed a significant decrease in cholesterol levels ($P < 0.05$). Hypercholesterolemia was associated with hypertriglyceridemia as compared with control animals. Hypertriglyceridemia also was significantly prevented by the treatment with *E. littorale*. There was no significant change in cholesterol and triglyceride levels in non-diabetic control rats treated with *E. littorale* as compared with non-diabetic control group (Table 2).

NIDDM control rats showed a significant increase ($P < 0.05$) in creatinine levels as compared with control animals. Treatment with *E. littorale* significantly decreased these values. There was also a significant increase in urea and BUN levels of NIDDM control rats as compared with control animals. *E. littorale* did not show any significant change in urea and BUN levels as compared with diabetic controls.

4. Discussion

In the present study, NIDDM control rats showed significantly higher levels of fasting and fed glucose levels as compared with non-diabetic control rats. This is consistent with earlier reports (Weir, et al., 1981; Gokhale et al., 1998). Hyperinsulinemia with low hepatic excretion and hypersecretion of beta cells are also reported in mild glucose intolerant obese subjects (Bonora et al., 1983). We also found increase in insulin levels and AUC_{insulin} after glucose load in neonatal STZ-diabetic rats. The high insulin concentration found in neonatal STZ-diabetic rats need not be of pancreatic origin. It could be due to metabolic alterations at extra pancreatic levels. In these rats, the metabolic clearance rate of insulin might have been altered. Insulin degradation following hormone receptor binding (Gliemann and Sonne, 1978) and reduced binding of insulin to its receptor have been reported in mild glucose intolerance (Olefsky, 1981). Therefore, the hyperinsulinemia in neonatal STZ-diabetic rats could be due to either decreased hepatic clearance of insulin or decreased number of insulin receptors, resulting in decreased

Table 1
Effect on General parameters of *E. littorale* in experimental animals ($n = 5-8$)

Parameters	Wistar control	Wistar treated with <i>E. littorale</i>	NIDDM control	NIDDM treated with <i>E. littorale</i>
Body weight (g)	228.3 $\pm$ 11.14	240.0 $\pm$ 6.12	160.0 $\pm$ 8.5 ^a	186.0 $\pm$ 6.3 ^b
Food intake (g per rat per day)	25 $\pm$ 3.2	22.0 $\pm$ 2.2	26.6 $\pm$ 1.2	23.3 $\pm$ 2.2
Water intake (ml per rat per day)	25.0 $\pm$ 3.5	26.0 $\pm$ 2.2	33.1 $\pm$ 2.5	29.0 $\pm$ 3.53

Each value is mean $\pm$ S.E.M.

^a Significantly different from Wistar control group ($P < 0.05$).

^b Significantly different from NIDDM control group ($P < 0.05$).

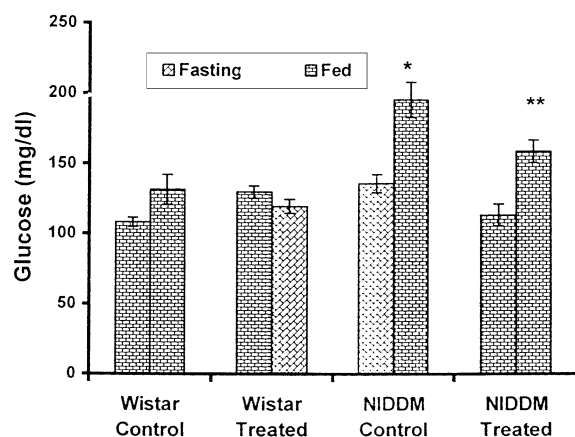


Fig. 1. Effect of losartan on serum glucose levels in NIDDM diabetic rats. Each bar depicts mean $\pm$ S.E.M. of five to eight animals. *Significantly different from control ($P < 0.05$). **Significantly different from NIDDM control ($P < 0.05$).

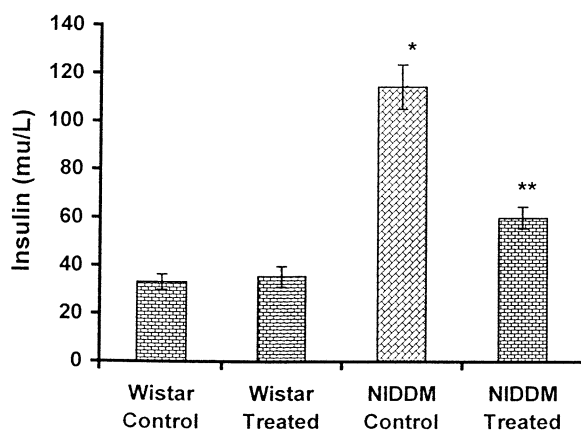


Fig. 2. Effect of losartan on serum insulin levels in NIDDM diabetic rats. Each bar depicts mean $\pm$ S.E.M. of five to eight animals. *Significantly different from control ($P < 0.05$). **Significantly different from NIDDM control ($P < 0.05$).

insulin binding and lowered insulin degradation. When animals were subjected to OGTT, the AUC_{insulin} of NIDDM control rats was significantly greater as compared with the non-diabetic rats. Despite high insulin levels, the AUC_{glucose} of the diabetic animals was greater than non-diabetic rats. However, AUC_{insulin} of NIDDM rats treated with *E. littorale* was found to be

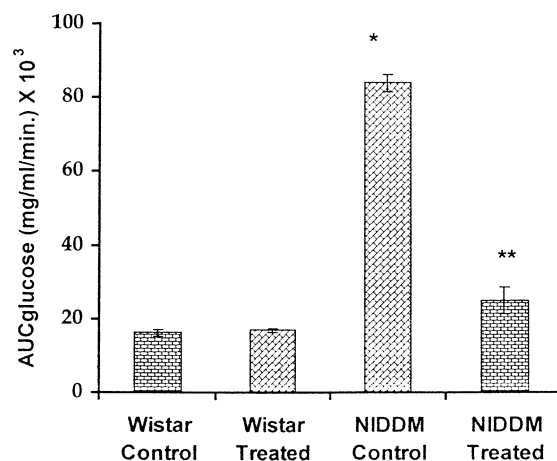


Fig. 3. Effect of losartan on AUC_{glucose} values in NIDDM diabetic rats. Each bar depicts mean $\pm$ S.E.M. of five to eight animals. *Significantly different from control ($P < 0.05$). **Significantly different from NIDDM control ($P < 0.05$).

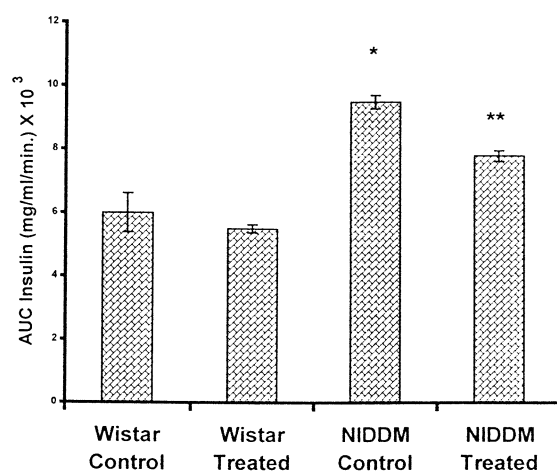


Fig. 4. Effect of losartan on AUC_{insulin} values in NIDDM diabetic rats. Each bar depicts mean $\pm$ S.E.M. of five to eight animals. *Significantly different from control ($P < 0.05$). **Significantly different from NIDDM control ($P < 0.05$).

significantly lower as compared with NIDDM control rats. This suggests that in normal animals *E. littorale* does not alter the release of insulin, but in conditions like hyperinsulinemia, it increases the insulin sensitivity for effective glucose disposal.

Table 2

Effect of *E. littorale* treatment on various biochemical parameters in Wistar and NIDDM rats ($n = 5-8$)

Parameters	Wistar control	Wistar treated with <i>E. littorale</i>	NIDDM control	NIDDM treated with <i>E. littorale</i>
K_{ITT} (per min)	8.41 ± 0.14	6.9 ± 2.0	5.28 ± 0.32^a	7.28 ± 0.38^b
$T_{1/2}$ glucose	8.24 ± 0.64	10.1 ± 0.35	13.13 ± 0.81^a	9.52 ± 0.1^b
Serum cholesterol (mg/dl)	75.22 ± 3.4	81.0 ± 2.8	93.7 ± 7.3^a	82.5 ± 8.4^b
Serum triglyceride (mg/dl)	83.34 ± 16.52	90.4 ± 11.14	162.5 ± 11.3^a	100.8 ± 13.5^b
Serum creatinine (mg/dl)	1.2 ± 0.05	1.74 ± 0.5	3.1 ± 0.4^a	2.1 ± 0.11^b
Serum urea (mg/dl)	22.1 ± 1.3	22.6 ± 2.4	33.7 ± 2.4^a	27.6 ± 1.7
Serum blood urea nitrogen (mg/dl)	10.3 ± 0.6	10.5 ± 1.13	15.7 ± 1.11^a	12.9 ± 0.8

Each value is mean $\pm$ S.E.M.^a Significantly different from Wistar control group ($P < 0.05$).^b Significantly different from NIDDM control group ($P < 0.05$).

The K_{ITT} was found to be significantly lower in neonatal STZ-diabetic rats as compared with controls. This indicates that NIDDM rats are insulin resistant. The specific mechanisms underlying in insulin resistant states are heterogeneous and may include a receptor defect (decrease in insulin sensitivity) or post receptor defect (decrease in responsiveness to insulin) or combination of both (Kahn, 1978; Crettaz and Jeanrenand, 1980). *E. littorale* treatment significantly increased K_{ITT} values.

ITT which represents the response to exogenously administered insulin on blood glucose has been used to estimate insulin sensitivity (Alford et al., 1971). ITT is a simple, reasonably accurate and rapid method for screening IR (Gruet et al., 1993). ITT indicates net result of resistance to insulin action at target level including receptor and post receptor defects. In the present investigation the rate of glucose disposal was found to be significantly decreased in NIDDM control rats as compared with Wistar control rats.

As reported earlier (Durrington, 1993) IR or insulin deficiency is associated with hypercholesterolemia and hypertriglyceridemia. STZ-diabetes showed increased plasma levels of cholesterol, triglyceride, free fatty acid and phospholipids (Rodrigues et al., 1986). Insulin deficiency or IR may be responsible for dyslipidemia, because insulin has an inhibitory action on HMG-COA reductase, a key rate-limiting enzyme responsible for the metabolism of cholesterol rich LDL particles. The mechanisms responsible for the development of hypertriglyceridemia in uncontrolled diabetes in humans (possibly in insulin deficient STZ-diabetic rats) are due to a number of metabolic abnormalities that occur sequentially. Acute insulin deficiency initially causes an increase in free fatty acid mobilization from adipose tissue, resulting in increased secretion of VLDL-triglyceride from liver (Balasse et al., 1972). With longer insulin deficiency liver converts free fatty acids into ketone bodies and VLDL-triglyceride secretion diminishes (Basso and Havel, 1970). At the same time, lipoprotein lipase activity falls (Nikkila et al., 1977)

resulting in impaired clearance of VLDL and chylomicrons from plasma (Bagdade et al., 1968). Reaven (1988) proposed that IR in diabetic (or non diabetic) subjects led to compensatory hyperinsulinemia, which is associated with, increased LDL and reduced HDL concentrations. In our study also NIDDM rats showed hypercholesterolemia and hypertriglyceridemia and the treatment with *E. littorale* significantly decreased both cholesterol and triglyceride levels. These findings also support the hypothesis that *E. littorale* causes improvement in insulin sensitivity.

In conclusion our data suggest that *E. littorale* aqueous extract in doses of 2 g/kg has potential antidiabetic activity. In addition to decreasing in serum glucose and lipids, it can also preserve kidney dysfunction in diabetic rats. This effect may be due to improvement in insulin sensitivity. Based on these beneficial effects *E. littorale* may be considered as supplementary herbal therapy in diabetic patients to prevent long term complications of diabetes mellitus.

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Update

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ELSEVIER



Corrigendum

Corrigendum to “Effect of chronic treatment with *Enicostemma littorale* in non-insulin-dependent diabetic (NIDDM) rats” [Journal of Ethnopharmacology 81 (2002) 199–204][☆]

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The authors regret that in the above published article the legends of Figs. 1–4 were incorrect. Losartan was stated instead of *E. littorale*. The correct legends are given below:

Fig. 1. Effect of *E. littorale* on serum glucose levels in NIDDM diabetic rats. Each bar depicts mean $\pm$ S.E.M. of 5–8 animals. (*) Significantly different from control ($P < 0.05$). (**) Significantly different from NIDDM control ($P < 0.05$).

Fig. 2. Effect of *E. littorale* on serum insulin levels in NIDDM diabetic rats. Each bar depicts mean $\pm$ S.E.M. of 5–8 animals. (*) Significantly different from control ($P < 0.05$). (**) Significantly different from NIDDM control ($P < 0.05$).

Fig. 3. Effect of *E. littorale* on AUC_{glucose} values in NIDDM diabetic rats. Each bar depicts mean $\pm$ S.E.M. of 5–8 animals. (*) Significantly different from control ($P < 0.05$). (**) Significantly different from NIDDM control ($P < 0.05$).

Fig. 4. Effect of *E. littorale* on AUC_{insulin} values in NIDDM diabetic rats. Each bar depicts mean $\pm$ S.E.M. of 5–8 animals. (*) Significantly different from control ($P < 0.05$). (**) Significantly different from NIDDM control ($P < 0.05$).

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